



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE

STA1023/MAT 1063- TUTORIAL 2

1. A certain town with a population of 100,000 has 3 newspapers: I, II, and III. The proportions of townspeople who read these papers are as follows:
I: 10 percent, I and II: 8 percent, I and II and III: 1 percent,
II: 30 percent, I and III: 2 percent,
III: 5 percent, II and III: 4 percent.
(The list tells us, for instance, that 8000 people read newspapers I and II.)
 - a) Find the number of people who read only one newspaper.
 - b) How many people read at least two newspapers?
 - c) If I and III are morning papers and II is an evening paper, how many people read at least one morning paper plus an evening paper?
 - d) How many people do not read any newspapers?
 - e) How many people read only one morning paper and one evening paper?
2. Two symmetric dice have had two of their sides painted red, two painted black, one painted yellow, and the other painted white. When this pair of dice is flipped, what is the probability that both dice land on the same color?
3. An urn contains 5 red, 6 blue, and 8 green balls. If a set of 3 balls is randomly selected, what is the probability that each of the balls will be
 - a) of the same color?
 - b) of different colors?
 - c) Repeat (a) and (b) under the assumption that whenever a ball is selected, its color is noted and it is then replaced in the urn before the next selection. (i.e. *sampling with replacement*)
4. In a state lottery, a player must choose 8 of the numbers from 1 to 40. The lottery commission then performs an experiment that selects 8 of these 40 numbers. Assuming that the choice of the lottery

commission is equally likely to be any of the $\binom{40}{8}$ combinations, what is the probability that a player has

- a) all 8 of the numbers selected by the lottery commission?
 - b) 7 of the numbers selected by the lottery commission?
 - c) at least 6 of the numbers selected by the lottery commission?
5. In answering a question on a multiple-choice test, a student either knows the answer or guesses. Let p be the probability that the student knows the answer and $(1 - p)$ be the probability that the student guesses. Assume that a student who guesses at the answer will be correct with probability $1/m$, where m is the number of multiple-choice alternatives. What is the conditional probability that a student knew the answer to a question given that he or she answered it correctly?
6. An insurance company believes that people can be divided into two classes: those who are accident prone and those who are not. The company's statistics show that an accident-prone person will have an accident at some time within a fixed 1-year period with probability 0.4, whereas this probability decreases to 0.2 for a non-accident-prone person. If we assume that 30 percent of the population is accident prone,
- a) what is the probability that a new policyholder will have an accident within a year of purchasing a policy?
 - b) Suppose that a new policyholder has an accident within a year of purchasing a policy. What is the probability that he or she is accident prone?
7. A bin contains 3 types of disposable flashlights. The probability that a type 1 flashlight will give more than 100 hours of use is 0.7, with the corresponding probabilities for type 2 and type 3 flashlights being 0.4 and 0.3, respectively. Suppose that 20 percent of the flashlights in the bin are type 1, 30 percent are type 2, and others are type 3.
- a). What is the probability that a randomly chosen flashlight will give more than 100 hours of use?
 - b) Given that a flashlight lasted more than 100 hours, what is the conditional probability that it was a type j flashlight, $j = 1, 2, 3$?

Submit on or before 23rd of July, 2026 by 4.00 pm.